

INTERNSHIP PROJECT

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IoT Based Weather Monitoring System

Using Arduino and Breadboard Simulation in Tinkercad

A Technical Project Report

Domain: Internet of Things (IoT) / Embedded Systems

Tools Used: Arduino UNO, DHT11/DHT22, Tinkercad

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1. Introduction

Weather conditions play a critical role in agriculture, transportation, disaster management, and daily human activity. Traditional weather stations are often bulky, expensive, and located far from the point of use, making real-time, localized monitoring difficult for individuals, small farms, or research setups. The Internet of Things (IoT) offers a low-cost and scalable alternative by connecting simple sensors to microcontrollers that can measure, process, and transmit environmental data continuously.

This report presents the design and simulation of an IoT Based Weather Monitoring System built around the Arduino UNO microcontroller. The system reads temperature, humidity, and other environmental parameters using sensors, displays the data locally, and can be extended to upload readings to the cloud for remote access. To validate the design before physical assembly, the complete circuit was built and simulated on a virtual breadboard using Tinkercad Circuits, an online platform that allows electronics prototyping and Arduino code testing without any physical hardware.

Using Tinkercad allows the design to be tested for wiring errors, sensor logic, and code correctness in a risk-free environment, which is especially useful for students and hobbyists before investing in physical components.

2. Objectives

- 1. To design a low-cost weather monitoring system using Arduino UNO.
- 2. To interface a DHT11/DHT22 sensor for measuring temperature and humidity.
- 3. To build and test the complete circuit virtually using Tinkercad's breadboard simulator.
- 4. To display real-time sensor readings on a 16x2 LCD module.
- 5. To write and simulate Arduino firmware (C/C++) that reads sensor data and controls the display.
- 6. To lay the groundwork for future IoT connectivity (Wi-Fi/cloud upload) using modules such as the ESP8266.

3. Components Required

The following components were used to design and simulate the circuit in Tinkercad:

Component	Quantity	Purpose
Arduino UNO R3	1	Main microcontroller board that processes sensor data
DHT11 / DHT22 Sensor	1	Measures ambient temperature and humidity
16x2 LCD Display	1	Displays real-time readings
Potentiometer (10k ohm)	1	Adjusts LCD contrast
Breadboard	1	Solderless platform for prototyping connections
Jumper Wires	As required	Connects components on the breadboard
Photoresistor (LDR)	1 (optional)	Measures ambient light intensity

Component	Quantity	Purpose
Push Button	1 (optional)	Manual refresh / mode-change input
Resistors (220 ohm, 10k ohm)	As required	Current limiting and pull-down for LDR/LED
USB Cable	1	Powers the Arduino and uploads code (in Tinkercad this is simulated)

4. System Architecture and Working Principle

The system follows a simple sense-process-display architecture:

7. Sensing: The DHT11/DHT22 sensor continuously measures temperature and humidity from the surrounding environment and sends the data as a digital signal to the Arduino.
8. Processing: The Arduino UNO reads the incoming signal using a dedicated sensor library, converts it into usable temperature (°C) and humidity (%) values, and checks the readings at fixed intervals (e.g., every 2 seconds).
9. Display: The processed values are formatted and shown on the 16x2 LCD screen so the user can view live weather data on-site.
10. (Optional) Transmission: For a full IoT deployment, an ESP8266 Wi-Fi module or similar can be added so that readings are also pushed to a cloud dashboard (e.g., ThingSpeak or Blynk) for remote monitoring and historical logging.

This layered approach keeps the core logic simple while allowing the system to be extended with additional sensors (rain, pressure, light) or connectivity modules without redesigning the base circuit.

5. Circuit Design and Simulation in Tinkercad

Tinkercad Circuits was used to design and validate the hardware connections before any physical build. The steps followed were:

11. Created a new circuit workspace in Tinkercad and placed an Arduino UNO board and a full-size breadboard on the canvas.
12. Placed the DHT11 sensor on the breadboard and connected its VCC pin to the Arduino 5V rail, GND to the Arduino GND rail, and the data pin to digital pin 2 of the Arduino (with a 10k ohm pull-up resistor between VCC and the data line, as per DHT sensor requirements).
13. Mounted the 16x2 LCD on the breadboard and wired it in 4-bit mode: RS, Enable, and data pins D4-D7 connected to Arduino digital pins, with the potentiometer wired between 5V and GND, its wiper connected to the LCD contrast (V0) pin.
14. Connected the LCD backlight pins (A and K) through a 220 ohm resistor to prevent excessive current draw.
15. Verified every connection visually using Tinkercad's wire-highlight feature, and used the built-in code editor to write and simulate the Arduino sketch directly on the virtual board.
16. Ran the simulation in real time, adjusting the DHT11's simulated temperature/humidity slider in Tinkercad to confirm the LCD updated correctly.

5.1 Breadboard Layout Summary

Arduino Pin	Connected To
5V	Breadboard + rail (powers DHT11 and LCD)

Arduino Pin	Connected To
GND	Breadboard - rail (common ground)
Digital Pin 2	DHT11 Data pin
Digital Pins 7, 8	LCD RS, Enable
Digital Pins 9, 10, 11, 12	LCD D4, D5, D6, D7
Analog A0 (optional)	LDR light sensor voltage divider

This wiring table mirrors the standard Arduino LiquidCrystal library pin mapping and the standard DHT11 single-wire digital interface, both of which are directly supported in Tinkercad's component library.

6. Arduino Program (Firmware)

The firmware was written in the Arduino IDE syntax (C/C++) directly inside Tinkercad's code editor, using the DHT sensor library and the LiquidCrystal library. At a high level, the program performs the following steps in the loop() function: it reads humidity and temperature from the DHT11, checks that the reading is valid (not NaN), clears and updates the LCD with formatted text, and then waits briefly before repeating the cycle.

Core logic implemented in the sketch:

- Include DHT.h and LiquidCrystal.h libraries and define sensor pin and type.
- In setup(): initialize the LCD with lcd.begin(16, 2) and start the sensor with dht.begin().
- In loop(): call dht.readTemperature() and dht.readHumidity(), print labels and values to the LCD using lcd.print(), and insert a delay of about 2000 milliseconds between readings, matching the DHT11's minimum sampling interval.
- Add basic error handling to display "Sensor Error" on the LCD if a reading returns an invalid value.

This structure keeps the firmware compact and easy to extend — for example, adding Serial.print() statements to send the same data to a Wi-Fi module for cloud logging.

7. Results and Observations

The simulated circuit was tested by running the Tinkercad simulation and adjusting the DHT11 sensor's virtual environment sliders to mimic changing weather conditions. The following observations were recorded:

Test Case	Simulated Input	LCD Output
Normal room condition	25°C, 50% RH	Temp: 25 C Hum: 50%
Hot condition	38°C, 30% RH	Temp: 38 C Hum: 30%
Humid condition	22°C, 85% RH	Temp: 22 C Hum: 85%
Sensor disconnected	No signal	Sensor Error

The LCD updated correctly within 2-3 seconds of each simulated change, confirming that the sensor-reading logic, formatting, and display refresh cycle functioned as intended. No wiring conflicts or code compilation errors were observed during simulation, indicating the design is ready for physical hardware assembly.

8. Applications

- Smart agriculture: monitoring field temperature and humidity to guide irrigation decisions.
- Home automation: triggering fans, humidifiers, or alerts based on indoor climate conditions.
- Weather stations for schools and hobby projects to teach IoT and embedded systems concepts.
- Greenhouse climate control and early warning systems for extreme conditions.
- Environmental data logging for research and citizen-science projects.

9. Advantages and Limitations

9.1 Advantages

- Low-cost components and open-source Arduino ecosystem.
- Tinkercad simulation eliminates the risk of damaging real components while learning.
- Simple to extend with additional sensors or a Wi-Fi module for full IoT capability.
- Real-time local display without needing constant internet access.

9.2 Limitations

- DHT11 has limited accuracy ($\pm 2^{\circ}\text{C}$, $\pm 5\%$ RH) compared to industrial-grade sensors.
- Tinkercad simulation does not perfectly replicate real-world sensor noise or timing delays.
- Without an added Wi-Fi/cloud module, the base design only supports local, on-site monitoring.

10. Conclusion and Future Scope

The IoT Based Weather Monitoring System designed and simulated in this project successfully demonstrates how an Arduino UNO, a DHT11 sensor, and an LCD display can be combined to measure and present real-time environmental data. Building and testing the circuit virtually in Tinkercad allowed all wiring and code to be verified before any physical prototyping, reducing the chance of hardware errors and component damage.

As future scope, the system can be enhanced by integrating an ESP8266 or ESP32 Wi-Fi module to push sensor data to a cloud platform such as ThingSpeak, Blynk, or Firebase, enabling remote access, historical data logging, graphing, and mobile alerts. Additional sensors such as a BMP180 (pressure), rain sensor, and anemometer could be added to build a more complete weather station, and solar power with a battery could make the unit fully field-deployable.

11. References

1. Arduino Official Documentation - <https://docs.arduino.cc>
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3. Adafruit DHT Sensor Library Documentation - <https://github.com/adafruit/DHT-sensor-library>
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